



SOMAIYA
VIDYAVIHAR UNIVERSITY

26.11.2025 (E)

Semester: November 2025– December 2025		
Maximum Marks: 50	Examination: End-Semester Examination	Duration: 2 Hrs.
Programme code: 27	Class: TY	Semester: V (SVU 2023)
Programme: B.Tech., Honors- Data Science for Emerging Analytics,(CCE, EXTC)		
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department/Section/Center: EXTC	
Course Code: 216h27C501	Name of the Course: Data Mining & Warehousing	
Instructions: 1) Draw neat diagrams wherever required 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any TWO	10
A	Describe the key features of a Data Warehouse.	05
B	Perform a tabular comparison of OLTP and OLAP systems.	05
C	What is a data cube in data warehouse conceptual model? Explain with reference to concept hierarchies.	05
Q.2	Attempt any ONE	10
A	Illustrate with a sketch the three-tier architecture of a data warehouse system.	10
B	Consider a data warehouse that is being created for an educational Institute. The following information is given to design a data warehouse scheme: Number of dimensions: 4 List of dimensions: student, instructor, course, semester Measures: count, average grade. At the lower conceptual level, the average grade measure stores the actual course grade of the student whereas at higher conceptual levels, average grade stores the average grade for the given combination. Perform the following operations: i. Design a star schema diagram of the data warehouse. ii. Starting from the base cuboid [student, instructor, course, semester], what specific OLAP operations should be performed to list the average grade of CS courses for each student. iii. If each dimension has five levels, then how many cuboids will this cube contain including the base and apex cuboid.	10
Q.3	Answer the following	10
A	Differentiate between Bagging and Boosting techniques.	5
B	What are the parameters and hyper-parameters of the following Machine learning algorithms: Simple Linear Regression Polynomial Regression K-Nearest Neighbor Describe how do we obtain them from the given dataset.	5



Q.4	Answer the following	10																																				
	Consider the table below, where the $(i, j)^{th}$ element of the table is the distance between data points x_i and x_j. Perform Complete linkage hierarchical clustering on data points, x_1, x_2, x_3, x_4, x_5. Sketch Dendrogram marking distance/height and data points. Determine the clusters of the data points obtained if Dendrogram is cut at the distance of i) 1, ii) 6 and iii) 10, respectively. <table><tr><th></th><th>x_1</th><th>x_2</th><th>x_3</th><th>x_4</th><th>x_5</th></tr><tr><th>x_1</th><td>0</td><td>9</td><td>3</td><td>6</td><td>11</td></tr><tr><th>x_2</th><td>9</td><td>0</td><td>7</td><td>5</td><td>10</td></tr><tr><th>x_3</th><td>3</td><td>7</td><td>0</td><td>9</td><td>2</td></tr><tr><th>x_4</th><td>6</td><td>5</td><td>9</td><td>0</td><td>8</td></tr><tr><th>x_5</th><td>11</td><td>10</td><td>2</td><td>8</td><td>0</td></tr></table>		x_1	x_2	x_3	x_4	x_5	x_1	0	9	3	6	11	x_2	9	0	7	5	10	x_3	3	7	0	9	2	x_4	6	5	9	0	8	x_5	11	10	2	8	0	10
	x_1	x_2	x_3	x_4	x_5																																	
x_1	0	9	3	6	11																																	
x_2	9	0	7	5	10																																	
x_3	3	7	0	9	2																																	
x_4	6	5	9	0	8																																	
x_5	11	10	2	8	0																																	
Q.5	Answer any ONE of the following	10																																				
A	Write a detailed note on Web Content , Web Structure & Web Usage Mining. State the applications of Web mining along with the mining tools used.	10																																				
B	Apply the Apriori algorithm on the following grocery store transaction database with support threshold $s=40\%$ and confidence threshold $c=60\%$. Show the candidate and frequent itemsets for each database scan. Enumerate all the final frequent itemsets. Generate the association rules from the mined frequent itemsets. Sort the association rules in the decreasing order of confidence																																					
	<table><tr><td>T1</td><td>Burger, Bread, Sauce</td></tr><tr><td>T2</td><td>Burger, Bread</td></tr><tr><td>T3</td><td>Burger, Cake, Chocolate</td></tr><tr><td>T4</td><td>Chocolate, Cake</td></tr><tr><td>T5</td><td>Chocolate, Sauce</td></tr></table>	T1	Burger, Bread, Sauce	T2	Burger, Bread	T3	Burger, Cake, Chocolate	T4	Chocolate, Cake	T5	Chocolate, Sauce																											
T1	Burger, Bread, Sauce																																					
T2	Burger, Bread																																					
T3	Burger, Cake, Chocolate																																					
T4	Chocolate, Cake																																					
T5	Chocolate, Sauce																																					